Linear equations in one variable — Part 2 answer key

#	ID	Answer	Difficulty
2.1	7084509a-2eca-4996-a36e-1e7ed19fdd12	C	Easy
2.2	027085-DC	?	Easy
2.3	f3aafaa2-b75f-4b93-a443-b88b79f2a8b1	B	Easy
2.4	06911-DC	D	Easy
2.5	55b379da-d6e2-4c9e-857d-8da822c1d1fa	130	Easy
2.6	4d9de101-04d3-4a9b-96ae-eef85f72bd23	A	Easy
2.7	ad73f33e-e2f7-442f-9463-281f925a0111	17	Easy
2.8	79f6861f-e1b2-4b3a-a56b-065a572689b2	A	Easy
2.9	85822077-c33d-45ef-bd16-1f8decf34dde	D	Medium
2.10	070610-DC	?	Medium
2.11	565e2580-0194-4be0-b5a7-5e2273ce4a14	C	Medium
2.12	657150dc-4949-4ed3-afe8-a00d56bf8642	C	Medium
2.13	11138-DC	B	Medium
2.14	028955-DC	B	Medium

2.15	ed19c902-acac-433c-b8f5-c8cd07c4e16d	D	Medium
2.16	8e8223b7-2824-48a9-a122-1abf78976190	A	Medium
2.17	027743-DC	A	Medium
2.18	a24605a1-dcc2-4d33-8f1f-a5e3f3904603	2	Medium
2.19	0964937e-bca9-49a1-ae3a-255505993ab8	C	Medium
2.20	05043-DC	?	Medium
2.21	08357-DC	A	Medium
2.22	08050-DC	B	Medium
2.23	81094fb4-28a4-4d27-b028-07b229772468	C	Medium
2.24	f0727bf9-4765-47a5-ad90-ee9cc0ed3794	C	Medium
2.25	ff27b3f5-a00f-4258-b4c6-8f41c7c06bf0	C	Medium
2.26	162bcb97-414c-46ed-a490-4d88402f07fa	C	Medium
2.27	028876-DC	B	Medium
2.28	4f88a160-8333-4005-8a92-1ba7b2c74b99	42	Medium
2.29	ba23c485-785e-419e-b55c-cb7482fba61e	60	Medium
2.30	022123-DC	D	Medium
2.31	568627eb-0784-4a63-a56a-ddffad314bb8	294	Medium
2.32	63a0e21a-3df7-463a-b0b7-6bf2eeefbcb4	C	Medium

2.33	6da250b9-a95c-48c8-b785-d54366f667e3	24.5, 49/2	Medium
2.34	d92e58bf-e6da-484f-8d84-5452a09619b1	D	Medium
2.35	070609-DC	?	Medium
2.36	4b647461-1e29-465f-a480-e7b4dcecc200	37	Medium
2.37	e2acc363-d0b7-4b19-ac43-34982f030a3d	B	Hard
2.38	10864-DC	D	Hard
2.39	c17521a8-2834-4b36-9f1a-dca9bf6e906d	-.9333, -14/15	Hard
2.40	98dc96db-8257-4f49-95af-515712d5e4e3	C	Hard
2.41	033d8e6a-f7e5-456a-b3ab-38b7ad87a27d	D	Hard
2.42	2a3b6ab6-2f25-4e2a-817b-9112e317c958	D	Hard
2.43	6d24c78b-537d-4e65-8a41-211301b6c972	A	Hard
2.44	9d5bc04d-81bd-4fda-8c78-8c26fa0de2f0	B	Hard
2.45	c5442207-cd62-4203-8c5a-95d42e060e41	B	Hard
2.46	7c776d67-dc96-4c86-97df-52e3204cdfe9	-.3266, -.3267, -49/150	Hard
2.47	d406296c-cba6-440d-94d6-5bf2accfba2a	-22	Hard
2.48	05397-DC	D	Hard

2.49	0797e1a5-ab62-4b2e-b8db-ec842244529a	11	Hard
2.50	08060-DC	C	Hard
2.51	ad6cdfd4-4643-45b2-b72d-0602fb30ce1c	403	Hard
2.52	830cf5a2-7b06-43e1-bf6c-fb7814a69db0	B	Hard
2.53	05364-DC	D	Hard
2.54	e9630635-49ff-4317-9dc7-81fb2b4d2e19	.9411, .9412, 16/17	Hard
2.55	eec9f1c3-9c60-420a-98bf-68fc90a14f95	B	Hard
2.56	0d482312-5261-43da-8a42-d4ab593d0949	D	Hard

Rationales

▼ 2.1 — C

Choice C is correct. Subtracting 6 from both sides of the given equation yields $4x = 12$, which is the equation given in choice C. Since this equation is equivalent to the given equation, it has the same solution as the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.2 — ?

The correct answer is 5. Subtracting $6x$ from both sides of the given equation gives $k = 5$, so for any value of x, $6x + k = 6x + 5$ if and only if $k = 5$. Therefore, if the given equation has infinitely many solutions, the value of k is 5.

▼ 2.3 — B

Choice B is correct. Multiplying both sides of the given equation by 4 yields $44x + 2 = 412$, or $16x + 8 = 48$. Therefore, the value of $16x + 8$ is 48.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.4 — D

Choice D is correct. The term 2n represents twice the number of CDs that Cathy has, and adding 3 represents 3 more than that amount.

Choices A and B are incorrect. The expression 3n represents three times the number of CDs that Cathy has. Choice C is incorrect. Subtracting 3 represents 3 fewer than twice the number of CDs that Cathy has.

▼ 2.5 — 130

The correct answer is 130. It's given that $8x - 7x + 130 = 260$. Combining like terms on the left-hand side of this equation yields $x + 130 = 260$. Subtracting 130 from each side of this equation yields $x = 130$. Therefore, the value of x that's the solution to the given equation is 130.

▼ 2.6 — A

Choice A is correct. It's given that p represents the number of pounds of strawberries Lorenzo purchased and Lorenzo paid $\$1.90$ per pound for the strawberries. It follows that the total amount, in dollars, Lorenzo paid for strawberries can be represented by $1.90p$. It's given that Lorenzo paid $\$2$ for the box of cereal. If Lorenzo paid a total of $\$9.60$ for the box of cereal and strawberries, it follows that the equation $1.90p + 2 = 9.60$ can be used to find p.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.7 — 17

The correct answer is 17. It's given that $2x + 3 = 9$. Multiplying each side of this equation by 3 yields $32x + 3 = 39$, or $6x + 9 = 27$. Subtracting 10 from each side of this equation yields $6x + 9 - 10 = 27 - 10$, or $6x - 1 = 17$. Therefore, the value of $6x - 1$ is 17.

▼ 2.8 — A

Choice A is correct. It's given that the rocket contained 467,000 kilograms (kg) of propellant before launch and had 362,105 kg remaining exactly 21 seconds after launch. Finding the difference between the amount, in kg, of propellant before launch and the remaining amount, in kg, of propellant after launch gives the amount, in kg, of propellant burned during the 21 seconds: $467{,}000 - 362{,}105 = 104{,}895$. Dividing the amount of propellant burned by the number of seconds yields $\frac{104{,}895}{21} = 4{,}995$. Thus, an average of 4,995 kg of propellant burned each second after launch.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from finding the amount of propellant burned, rather than the amount of propellant burned each second.

▼ 2.9 — D

Choice D is correct. It's given that the bowl starts with 20 ounces of water and has 9 ounces of water remaining after a period of time has passed. The amount of water the bowl has lost during the time period can be found by subtracting the remaining amount of water from the amount of water the bowl starts with, which yields $20 - 9$ ounces, or 11 ounces. This means the bowl loses 11 ounces of water during that period of time. It's given that the amount of water decreases by 1 ounce every 4 days. Letting t represent the number of days the bowl has been uncovered, it follows that $\frac{1}{4} = \frac{11}{t}$. Multiplying both sides of this equation by $4t$ yields $t = 44$. Therefore, the bowl has been uncovered for 44 days.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of t for the equation $\frac{1}{4} = \frac{9}{t}$, not $\frac{1}{4} = \frac{11}{t}$.

▼ 2.10 — ?

The correct answer is 1.2. One way to solve the equation $2(p+1)+8(p-1)=5p$ is to first distribute the terms outside the parentheses to the terms inside the parentheses: $2p+2+8p-8=5p$. Next, combine like terms on the left side of the equal sign: $10p-6=5p$. Subtracting 10p from both sides yields $-6=-5p$. Finally, dividing both sides by -5 gives $p=\frac{6}{5}$, which is equivalent to $p=1.2$. Note that 1.2 and 6/5 are examples of ways to enter a correct answer.

▼ 2.11 — C

Choice C is correct. If the two sides of a linear equation are equivalent, then the equation is true for any value. If an equation is true for any value, it has infinitely many solutions. Since the two sides of the given linear equation $66x = 66x$ are equivalent, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.12 — C

Choice C is correct. It's given that the cost to rent a bus is $\$950$ for the first 3 hours and an additional $\$50$ per hour for each hour after the first 3 hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first 3 hours can be represented by the expression $t - 3$. The cost for these additional hours is $\$50$ per hour, so the cost for the additional hours can be represented by the expression $50(t - 3)$. The total cost can be calculated by adding the cost for the first 3 hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 3)$. It's also given that the total cost to rent the bus from the company for t hours is $\$1{,}150$. Thus, the equation that represents this situation is $950 + 50(t - 3) = 1{,}150$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.13 — B

Choice B is correct. It is given that the width of the dance floor is w feet. The length is 6 feet longer than the width; therefore, the length of the dance floor is $w+6$. So the perimeter is $w+w+(w+6)+(w+6)=4w+12$.

Choice A is incorrect because it is the sum of one length and one width, which is only half the perimeter. Choice C is incorrect and may result from using the formula for the area instead of the formula for the perimeter and making a calculation error. Choice D is incorrect because this is the area, not the perimeter, of the dance floor.

▼ 2.14 — B

Choice B is correct. Because 2 is a factor of both $2x$ and 6, the expression $2x-6$ can be rewritten as $2(x-3)$. Substituting $2(x-3)$ for $(2x-6)$ on the left-hand side of the given equation yields $3(2)(x-3)-11=4(x-3)+6$, or $6(x-3)-11=4(x-3)+6$. Subtracting $4(x-3)$ from both sides of this equation yields $2(x-3)-11=6$. Adding 11 to both sides of this equation yields $2(x-3)=17$. Dividing both sides of this equation by 2 yields $x-3=\frac{17}{2}$

.

Alternate approach: Distributing 3 to the quantity $(2x-6)$ on the left-hand side of the given equation and distributing 4 to the quantity $(x-3)$ on the right-hand side yields $6x-18-11=4x-12+6$, or $6x-29=4x-6$. Subtracting $4x$ from both sides of this equation yields $2x-29=-6$. Adding 29 to both sides of this equation yields $2x=23$. Dividing both sides of this equation by 2 yields $x=\frac{23}{2}$. Therefore, the value of $x-3$ is $\frac{23}{2}-3$, or $\frac{17}{2}$.

Choice A is incorrect. This is the value of x, not $x-3$. Choices C and D are incorrect. If the value of $x-3$ is $\frac{15}{2}$ or $-\frac{15}{2}$, it follows that the value of x is $\frac{21}{2}$ or $-\frac{9}{2}$, respectively. However, solving the given equation for x yields $x=\frac{23}{2}$. Therefore, the value of $x-3$ can't be $\frac{15}{2}$ or $-\frac{15}{2}$.

▼ 2.15 — D

Choice D is correct. It's given that the candle starts with 17 ounces of wax and has 6 ounces of wax remaining after a period of time has passed. The amount of wax the candle has lost during the time period can be found by subtracting the remaining amount of wax from the amount of wax the candle was made of, which yields $17 - 6$ ounces, or 11 ounces. This means the candle loses 11 ounces of wax during that period of time. It's given that the amount of wax decreases by 1 ounce every 4 hours. If h represents the number of hours the candle has been burning, it follows that $\frac{1}{4} = \frac{11}{h}$. Multiplying both sides of this equation by $4h$ yields $h = 44$. Therefore, the candle has been burning for 44 hours.

Choice A is incorrect and may result from using the equation $\frac{1}{4} = \frac{h}{11}$ rather than $\frac{1}{4} = \frac{11}{h}$ to represent the situation, and then rounding to the nearest whole number.

Choice B is incorrect. This is the amount of wax, in ounces, remaining in the candle, not the number of hours it has been burning.

Choice C is incorrect and may result from using the equation $\frac{1}{4} = \frac{6}{h}$ rather than $\frac{1}{4} = \frac{11}{h}$ to represent the situation.

▼ 2.16 — A

Choice A is correct. It's given that the equation $15{,}000 = 2.00x - 4{,}500$ represents this situation for a month where x tape dispensers are created and sold. It's also given that the company calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. It follows that $2.00x$ represents the monthly sales revenue, in dollars. Therefore, the best interpretation of $2.00x$ in this context is the monthly sales revenue from selling x tape dispensers.

Choice B is incorrect. This is the best interpretation of 2.00, not $2.00x$.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of $4{,}500$, not $2.00x$.

▼ 2.17 — A

Choice A is correct. This equation has no solution when there is no value of x that produces a true statement. Solving the given equation for x by dividing both sides by $(b-2)$ gives

$x = \frac{8}{(b-2)}$. When $(b-2)=0$, the right-hand side of this equation will be undefined, and the equation will have no solution. Therefore, when $b=2$, there is no value of x that satisfies the given equation.

Choices B, C, and D are incorrect. Substituting 4, 6, and 10 for b in the given equation yields exactly one solution, rather than no solution, for x. For example, substituting 4 for b in the given equation yields $(4-2)x=8$, or $2x=8$. Dividing both sides of $2x=8$ by 2 yields $x=4$. Similarly, if $b=6$ or $b=10$, $x=2$ and $x=1$, respectively.

▼ 2.18 — 2

The correct answer is 2. An equation with one variable, x, has infinitely many solutions only when both sides of the equation are equal for any defined value of x. It's given that $2x + 16 = ax + 8$, where a is a constant. This equation can be rewritten as $2x + 8 = ax + 8$. If this equation has infinitely many solutions, then both sides of this equation are equal for any defined value of x. Both sides of this equation are equal for any defined value of x when $2 = a$. Therefore, if the equation has infinitely many solutions, the value of a is 2.

Alternate approach: If the given equation, $2x + 16 = ax + 8$, has infinitely many solutions, then both sides of this equation are equal for any value of x. If $x = 0$, then substituting 0 for x in $2x + 16 = ax + 8$ yields $20 + 16 = a0 + 8$, or $16 = 8a$. Dividing both sides of this equation by 8 yields $2 = a$.

▼ 2.19 — C

Choice C is correct. Subtracting 16 from both sides of the given equation yields $30 = 2(x - 8)$. Therefore, the value of $2(x - 8)$ is 30.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.20 — ?

Choice B is correct. Since p represents Megan’s regular pay per hour, 1.5p represents the pay per hour in excess of 8 hours. Since Megan worked for 10 hours, she must have been

paid p dollars per hour for 8 of the hours plus 1.5p dollars per hour for the remaining 2 hours. Therefore, since Megan earned $137.50 for the 10 hours, the situation can be represented by the equation 137.5 = 8p + 2(1.5)p. Distributing the 2 in the equation gives 137.5 = 8p + 3p, and combining like terms gives 137.5 = 11p. Dividing both sides by 11 gives p = 12.5. Therefore, Megan's regular wage is $12.50.

Choices A and C are incorrect and may be the result of calculation errors. Choice D is incorrect and may result from finding the average hourly wage that Megan earned for the 10 hours of work.

▼ 2.21 — A

Choice A is correct. Adding the like terms on the left-hand side of the given equation yields $5(x-5)=10$. Dividing both sides of this equation by 5 yields $x-5=2$.

Choice B is incorrect and may result from subtracting 5, not dividing by 5, on both sides of the equation $5(x-5)=10$. Choice C is incorrect. This is the value of x, not the value of $x-5$. Choice D is incorrect. This is the value of $x+5$, not the value of $x-5$.

▼ 2.22 — B

Choice B is correct. Distributing the a on the left-hand side of the equation gives 3a – b – ax = –1 – 2x. Rearranging the terms in each side of the equation yields –ax + 3a – b = –2x –1. Since the equation has infinitely many solutions, it follows that the coefficients of x and the free terms on both sides must be equal. That is, –a = –2, or a = 2, and 3a – b = –1. Substituting 2 for a in the equation 3a – b = –1 gives 3(2) – b = –1, so b = 7.

Choice A is incorrect and may be the result of a conceptual error when finding the value of b. Choices C and D are incorrect and may result from making a sign error when simplifying.

▼ 2.23 — C

Choice C is correct. It's given that the cost to rent a commercial fishing boat is $\$950$ for the first 2 hours and an additional $\$50$ per hour for each hour after the first 2 hours. It's also given that t represents the total number of hours and $t > 2$. Therefore, the number of additional hours after the first 2 hours can be represented with the expression $t - 2$. The cost for these additional hours is $\$50$ per hour, so the cost for the additional hours can be represented by the expression $50t - 2$. The total cost can be calculated by adding the cost for the first 2 hours to the cost for the additional hours and can be represented by the

expression $950 + 50t - 2$. It's also given that the total cost to rent the commercial fishing boat from the company for t hours is $\$ 1{,}100$. Thus, the equation that represents this situation is $950 + 50t - 2 = 1{,}100$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.24 — C

Choice C is correct. Dividing both sides of the given equation by 2 yields $5x = 3$. Therefore, the value of $5x$ is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.25 — C

Choice C is correct. Subtracting $4x + 4$ from both sides of the given equation yields $1x + 4 = 29$, or $x + 4 = 29$. Therefore, the value of $x + 4$ is 29.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not $x + 4$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.26 — C

Choice C is correct. If an equation has infinitely many solutions, then the two sides of the equation must be equivalent. Multiplying each side of the given equation by 2 yields $8x + 24 = ax + 3$. Since 8 is a common factor of both terms on the left-hand side of this equation, the equation can be rewritten as $8x + 3 = ax + 3$. The two sides of this equation are equivalent when $a = 8$. Therefore, if the given equation has infinitely many solutions, the value of a is 8.

Alternate approach: If the given equation, $4x + 12 = \frac{ax+3}{2}$, has infinitely many solutions, then both sides of this equation are equal for any value of x. If $x = 0$, then substituting 0 for x in the given equation yields $40 + 12 = \frac{a0+3}{2}$, or $12 = \frac{3}{2}a$. Dividing both sides of this equation by $\frac{3}{2}$ yields $8 = a$.

Choice A is incorrect. If the value of a is 0, the given equation is equivalent to $4x + 12 = 0$, which has one solution, not infinitely many solutions.

Choice B is incorrect. If the value of a is 3, the given equation is equivalent to $4x + 12 = \frac{3x+3}{2}$, or $4x + 12 = \frac{3}{2}x + \frac{9}{2}$, which has one solution, not infinitely many solutions.

Choice D is incorrect. If the value of a is 12, the given equation is equivalent to $4x + 12 = \frac{12x+3}{2}$, or $4x + 12 = 6x + 18$, which has one solution, not infinitely many solutions.

▼ 2.27 — B

Choice B is correct. The height of the tree at a given time is equal to its height when it was planted plus the number of feet that the tree grew. In the given equation, 14 represents the height of the tree at the given time, and 6 represents the height of the tree when it was planted. It follows that $2n$ represents the number of feet the tree grew from the time it was planted until the time it reached a height of 14 feet. Since n represents the number of years between the given time and the time the tree was planted, 2 must represent the average number of feet the tree grew each year.

Choice A is incorrect and may result from interpreting the coefficient 2 as doubling instead of as increasing by 2 each year. Choice C is incorrect. The height of the tree when it was 1 year old was $2(1)+6=8$ feet, not 2 feet. Choice D is incorrect. No information is given to connect the growth of one particular tree to the growth of similar trees.

▼ 2.28 — 42

The correct answer is 42. The expression $x + 6$ is a factor of both terms on the left-hand side of the given equation. Therefore, the given equation can be written as $x + 6\frac{1}{3} - \frac{1}{2} = -8$, or $x + 6\frac{1}{6} = -8$. Multiplying each side of this equation by -6 yields $x + 6 = 48$. Subtracting 6 from each side of this equation yields $x = 42$. Therefore, the value of x that is the solution to the given equation is 42.

▼ 2.29 — 60

The correct answer is 60. Subtracting t from both sides of the given equation yields $2(3t - 10) = 40 + 3t$. Applying the distributive property to the left-hand side of this equation yields $6t - 20 = 40 + 3t$. Adding 20 to both sides of this equation yields $6t = 60 + 3t$. Subtracting $3t$ from both sides of this equation yields $3t = 60$. Therefore, the value of $3t$ is 60.

▼ 2.30 — D

Choice D is correct. Multiplying the given equation by 2 on each side yields $2\left(4x-\frac{1}{2}\right)=2(-5)$. Applying the distributive property, this equation can be rewritten as $2(4x)-2\left(\frac{1}{2}\right)=2(-5)$, or $8x-1=-10$.

Choices A, B, and C are incorrect and may result from calculation errors in solving the given equation for x and then substituting that value of x in the expression $8x-1$.

▼ 2.31 — 294

The correct answer is 294. Subtracting 18 from each side of the given equation yields $\frac{6}{7}p = 36$. Multiplying each side of this equation by $\frac{7}{6}$ yields $p = 42$. Multiplying each side of this equation by 7 yields $7p = 294$. Therefore, the value of $7p$ is 294.

▼ 2.32 — C

Choice C is correct. For the given equation, $x + 5$ is a factor of both terms on the left-hand side. Therefore, the given equation can be rewritten as $\frac{1}{4} - \frac{1}{3}x + 5 = -7$, or $\frac{3}{12} - \frac{4}{12}x + 5 = -7$, which is equivalent to $-\frac{1}{12}x + 5 = -7$. Multiplying both sides of this equation by -12 yields $x + 5 = 84$. Subtracting 5 from both sides of this equation yields $x = 79$.

Choice A is incorrect. This is the value of x for which the left-hand side of the given equation equals $\frac{7}{12}$, not -7.

Choice B is incorrect. This is the value of x for which the left-hand side of the given equation equals 0, not -7.

Choice D is incorrect. This is the value of x for which the left-hand side of the given equation equals $-\frac{209}{12}$, not -7.

▼ 2.33 — 24.5, 49/2

The correct answer is $\frac{49}{2}$. It's given that $9r = 7(r + 7)$. Applying the distributive property on the right-hand side of this equation yields $9r = 7r + 49$. Subtracting $7r$ from both sides of this equation yields $2r = 49$. Dividing both sides of this equation by 2 yields $r = \frac{49}{2}$. Therefore, the solution to the given equation is $\frac{49}{2}$. Note that 49/2 and 24.5 are examples of ways to enter a correct answer.

▼ 2.34 — D

Choice D is correct. The value of $4 - 3x$ can be found by isolating this expression in the given equation. Subtracting 2 from both sides of the given equation yields $94 - 3x = 84 - 3x + 16$. Subtracting $84 - 3x$ from both sides of this equation yields $94 - 3x - 84 - 3x = 16$, which gives $14 - 3x = 16$, or $4 - 3x = 16$. Therefore, the value of $4 - 3x$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not $4 - 3x$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.35 — ?

The correct answer is 22.4. If the height of the plant increased by an average of 1.20 centimeters per day for 12 days, then its total growth over the 12 days was $(1.20)(12) = 14.4$ centimeters. The plant was 36.8 centimeters tall after 12 days, so at the beginning of the study its height was $36.8 - 14.4 = 22.4$ centimeters. Note that 22.4 and 112/5 are examples of ways to enter a correct answer.

Alternate approach: The equation $36.8 = 12(1.20) + h$ can be used to represent this situation, where h is the height of the plant, in centimeters, at the beginning of the study. Solving this equation for h yields 22.4 centimeters.

▼ 2.36 — 37

The correct answer is 37. It's given that the museum earns revenue of $\$ 14$ for each tablet rented for the day. It's also given that on Wednesday, the museum earned $\$ 406$ in profit from renting tablets after paying daily expenses of $\$ 112$. Let x represent the number of tablets the museum rented on Wednesday. It follows that the total revenue can be represented by the expression $14x$. Because $\text{profit} = \text{total revenue} - \text{total expenses}$, the

equation $406 = 14x - 112$ represents this situation. Adding 112 to both sides of this equation yields $14x = 518$. Dividing both sides of this equation by 14 yields $x = 37$. Therefore, the museum rented 37 tablets on Wednesday.

▼ 2.37 — B

Choice B is correct. It's given that each side of a 30-sided polygon has one of three lengths. It's also given that the number of sides with length 8 centimeters cm is 5 times the number of sides n with length $3\ \text{cm}$. Therefore, there are $5 \times n$, or $5n$, sides with length $8\ \text{cm}$. It's also given that there are 6 sides with length $4\ \text{cm}$. Therefore, the number of $3\ \text{cm}$, $4\ \text{cm}$, and $8\ \text{cm}$ sides are n, 6, and $5n$, respectively. Since there are a total of 30 sides, the equation $n + 6 + 5n = 30$ represents this situation. Combining like terms on the left-hand side of this equation yields $6n + 6 = 30$. Therefore, the equation that must be true for the value of n is $6n + 6 = 30$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.38 — D

Choice D is correct. For a linear equation in a form $ax + b = cx + d$ to have no solutions, the x-terms must have equal coefficients and the remaining terms must not be equal. Expanding the right-hand side of the given equation yields $9x + 5 = ax + ab$. Inspecting the x-terms, 9 must equal a, so statement I must be true. Inspecting the remaining terms, 5 can't equal $9b$. Dividing both of these quantities by 9 yields that b can't equal $\frac{5}{9}$. Therefore, statement III must be true. Since b can have any value other than $\frac{5}{9}$, statement II may or may not be true.

Choice A is incorrect. For the given equation to have no solution, both $a = 9$ and $b \neq \frac{5}{9}$ must be true. Choice B is incorrect because it must also be true that $b \neq \frac{5}{9}$. Choice C is incorrect because when $a = 9$, there are many values of b that lead to an equation having no solution. That is, b might be 5, but b isn't required to be 5.

▼ 2.39 — -.9333, -14/15

The correct answer is $-\frac{14}{15}$. A linear equation in the form $ax + b = cx + d$ has no solution only when the coefficients of x on each side of the equation are equal and the constant terms are not equal. Dividing both sides of the given equation by 2 yields $kx - n = -\frac{28}{30}x - \frac{36}{38}$, or $kx - n = -\frac{14}{15}x - \frac{18}{19}$. Since it's given that the equation has no solution, the coefficient of x on both sides of this equation must be equal, and the constant terms on both sides of this equation must not be equal. Since $\frac{18}{19} < 1$, and it's given that $n > 1$, the second condition is true. Thus, k must be equal to $-\frac{14}{15}$. Note that -14/15, -.9333, and -0.933 are examples of ways to enter a correct answer.

▼ 2.40 — C

Choice C is correct. Applying the distributive property to each side of the given equation yields $150x - 90 = -90 + 150x$. Applying the commutative property of addition to the right-hand side of this equation yields $150x - 90 = 150x - 90$. Since the two sides of the equation are equivalent, this equation is true for any value of x. Therefore, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.41 — D

Choice D is correct. It's given that during a certain day at a factory, the number of 7-inch concrete screws the factory makes is n and the number of 4-inch concrete screws the factory makes is 22. It's also given that during this day the number of 9-inch concrete screws the factory makes is 5 times the number of 7-inch concrete screws, or $5n$. Therefore, the total number of 7-inch, 9-inch, and 4-inch concrete screws is $n + 5n + 22$, or $6n + 22$. It's given that during this day, the factory makes 100 concrete screws total. Thus, the equation $6n + 22 = 100$ represents this situation.

Choice A is incorrect. This equation represents a situation where the total length, in inches, of all the concrete screws, rather than the total number of concrete screws, is 100.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the total number of 9-inch concrete screws and 4-inch concrete screws, not including the 7-inch concrete screws, is 100.

▼ 2.42 — D

Choice D is correct. In the given equation, $x(r - 9) + 4 = 19x + 27$, r is a positive integer. Distributing on the left-hand side of this equation yields $rx - 9x + 4 = 19x + 27$. Subtracting 4 from both sides of this equation yields $rx - 9x = 19x + 23$. Subtracting $19x$ from both sides of this equation yields $rx - 28x = 23$, which can be written as $(r - 28)x = 23$. For this equation to have exactly one solution, the coefficient of x, $(r - 28)$, cannot be equal to 0. Therefore, 28 CANNOT be the value of r.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.43 — A

Choice A is correct. Distributing 12 on the left-hand side and -3 on the right-hand side of the given equation yields $12x - 36 = -3x - 36$. Adding $3x$ to each side of this equation yields $15x - 36 = -36$. Adding 36 to each side of this equation yields $15x = 0$. Dividing each side of this equation by 15 yields $x = 0$. This means that 0 is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.44 — B

Choice B is correct. A linear equation in one variable has no solution if and only if the equation is false; that is, when there is no value of x that produces a true statement. It's given that in the equation $-3x + 21px = 84$, p is a constant and the equation has no solution for x. Therefore, the value of the constant p is one that results in a false equation. Factoring out the common factor of $-3x$ on the left-hand side of the given equation yields $-3x1 - 7p = 84$. Dividing both sides of this equation by -3 yields $x1 - 7p = -28$. Dividing both sides of this

equation by $1 - 7p$ yields $x = \frac{-28}{1 - 7p}$. This equation is false if and only if $1 - 7p = 0$. Adding $7p$ to both sides of $1 - 7p = 0$ yields $1 = 7p$. Dividing both sides of this equation by 7 yields $\frac{1}{7} = p$. It follows that the equation $x = \frac{-28}{1 - 7p}$ is false if and only if $p = \frac{1}{7}$. Therefore, the given equation has no solution if and only if the value of p is $\frac{1}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.45 — B

Choice B is correct. Adding $98x$ to each side of the given equation yields $49x = 0$. Dividing each side of this equation by 49 yields $x = 0$. This means that 0 is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.46 — -.3266, -.3267, -49/150

The correct answer is $-.3267$. Applying the distributive property to the right-hand side of the given equation yields $0.8t - 0.46 = 8t - 0.008 + 1.9$, or $0.8t - 0.46 = 8t + 1.892$. Subtracting $0.8t$ from both sides of this equation yields $-0.46 = 7.2t + 1.892$. Subtracting 1.892 from both sides of this equation yields $-2.352 = 7.2t$. Dividing both sides of this equation by 7.2 yields $\frac{-2.352}{7.2} = t$. Therefore, the value of t is approximately -0.32667. Note that -.3267, -.3266, -0.326, and -0.327 are examples of ways to enter a correct answer.

▼ 2.47 — -22

The correct answer is -22. The given equation can be rewritten as $-2t + 3 = 38$. Dividing both sides of this equation by -2 yields $t + 3 = -19$. Subtracting 3 from both sides of this equation yields $t = -22$. Therefore, -22 is the value of t that is the solution to the given equation.

▼ 2.48 — D

Choice D is correct. After using the auger for 5 hours, Hector had removed 24,000 – 19,350 = 4,650 bushels of corn from the storage bin. During the 5-hour period, the auger removed

corn from the bin at a constant rate of $\frac{4,650}{5} = 930$ bushels per hour. Assuming the auger continues to remove corn at this rate, after x hours it will have removed 930x bushels of corn. Because the bin contained 24,000 bushels of corn when Hector started using the auger, the equation 24,000 – 930x = 12,840 can be used to find the number of hours, x, Hector will have been using the auger when 12,840 bushels of corn remain in the bin. Subtracting 12,840 from both sides of this equation and adding 930x to both sides of the equation yields 11,160 = 930x. Dividing both sides of this equation by 930 yields x = 12. Therefore, Hector will have been using the auger for 12 hours when 12,840 bushels of corn remain in the storage bin.

Choice A is incorrect. Three hours after Hector began using the auger, 24,000 – 3(930) = 21,210 bushels of corn remained, not 12,840. Choice B is incorrect. Seven hours after Hector began using the auger, 24,000 – 7(930) = 17,490 bushels of corn will remain, not 12,840. Choice C is incorrect. Eight hours after Hector began using the auger, 24,000 – 8(930) = 16,560 bushels of corn will remain, not 12,840.

▼ 2.49 — 11

The correct answer is 11. Subtracting 5 from each side of the given equation yields $-72 - 4x = 11 - 82 - 4x$. Adding $82 - 4x$ to each side of this equation yields $2 - 4x = 11$. Therefore, the value of $2 - 4x$ is 11.

▼ 2.50 — C

Choice C is correct. It is given that x represents the number of stations that will be set up for Experiment A and that there will be 5 stations total, so it follows that 5 – x is the number of stations that will be set up for Experiment B. It is also given that Experiment A requires 6 teaspoons of salt and that Experiment B requires 4 teaspoons of salt, so the total number of teaspoons of salt required is 6x + 4(5 – x), which simplifies to 2x + 20.

Choices A, B, and D are incorrect and may be the result of not understanding the description of the context.

▼ 2.51 — 403

The correct answer is 403. For a linear equation in one variable to have infinitely many solutions, the coefficients of the variable must be equal on both sides of the equation and the constant terms must also be equal on both sides of the equation. The given equation can be rewritten as $\frac{43x+7}{4} - \frac{s}{13} = rx - 8$, or $3x + 7 - \frac{s}{13} = rx - 8$. Applying the distributive property

to the right-hand side of this equation yields $3x + 7 - \frac{s}{13} = rx - 8r$. For this equation to have infinitely many solutions, the coefficients of x must be equal, so it follows that $3 = r$. Additionally, the constant terms must be equal, which means $7 - \frac{s}{13} = -8r$. Substituting 3 for r in this equation yields $7 - \frac{s}{13} = -83$, or $7 - \frac{s}{13} = -24$. Adding $\frac{s}{13}$ to both sides of this equation yields $7 = -24 + \frac{s}{13}$. Adding 24 to both sides of this equation yields $31 = \frac{s}{13}$. Multiplying both sides of this equation by 13 yields $403 = s$. Therefore, if the equation has infinitely many solutions, the value of s is 403.

▼ 2.52 — B

Choice B is correct. Multiplying both sides of the given equation by 313, or 39, yields $39\frac{x+6}{3} = 39\frac{x+6}{13}$, or $13x + 6 = 3x + 6$. Subtracting $3x + 6$ from both sides of this equation yields $10x + 6 = 0$. Dividing both sides of this equation by 10 yields $x + 6 = 0$. Therefore, if $\frac{x+6}{3} = \frac{x+6}{13}$, then the value of $x + 6$ is 0. It follows that of the given choices, the value of $x + 6$ is between -2 and 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.53 — D

Choice D is correct. Since gasoline costs $4 per gallon, and since Alan's car travels an average of 25 miles per gallon, the expression $\frac{4}{25}$ gives the cost, in dollars per mile, to drive the car. Multiplying $\frac{4}{25}$ by m gives the cost for Alan to drive m miles in his car. Alan wants to reduce his weekly spending by $5, so setting $\frac{4}{25}$m equal to 5 gives the number of miles, m, by which he must reduce his driving.

Choices A, B, and C are incorrect. Choices A and B transpose the numerator and the denominator in the fraction. The fraction $\frac{25}{4}$ would result in the unit miles per dollar, but the question requires a unit of dollars per mile. Choices A and C set the expression equal to 95 instead of 5, a mistake that may result from a misconception that Alan wants to reduce his driving by 5 miles each week; instead, the question says he wants to reduce his weekly expenditure by $5.

▼ 2.54 — .9411, .9412, 16/17

The correct answer is $\frac{16}{17}$. It's given that the equation $3kx + 13 = \frac{48}{17}x + 36$ has no solution. A linear equation in the form $ax + b = cx + d$, where a, b, c, and d are constants, has no solution only when the coefficients of x on each side of the equation are equal and the constant terms aren't equal. Dividing both sides of the given equation by 3 yields $kx + 13 = \frac{48}{51}x + \frac{36}{3}$, or $kx + 13 = \frac{16}{17}x + 12$. Since the coefficients of x on each side of the equation must be equal, it follows that the value of k is $\frac{16}{17}$. Note that 16/17, .9411, .9412, and 0.941 are examples of ways to enter a correct answer.

▼ 2.55 — B

Choice B is correct. Multiplying both sides of the given equation by $(7)(9)$, or 63, yields $(63)\frac{x-5}{7} = (63)\frac{x-5}{9}$, or $9(x - 5) = 7(x - 5)$. Subtracting $7(x - 5)$ from both sides of this equation yields $2(x - 5) = 0$. Dividing both sides of this equation by 2 yields $x - 5 = 0$. Therefore, if $\frac{x-5}{7} = \frac{x-5}{9}$, then the value of $x - 5$ is 0. It follows that of the given choices, the value of $x - 5$ is between -3 and 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.56 — D

Choice D is correct. It's given that during a certain day, the number of 9-inch frying pans the manufacturing plant makes is n and the number of 7-inch frying pans it makes is 10. It's also given that during this day the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number of 9-inch frying pans, or $4n$. Therefore, the total number of 7-inch, 9-inch, and 10-inch frying pans the manufacturing plant makes is $n + 10 + 4n$, or $5n + 10$. It's given that during this day the manufacturing plant makes 100 frying pans total. Thus, the equation $5n + 10 = 100$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.